

CHAPTER 9

Budgets

9.1. Introduction

This chapter describes pointing, propellant, and mass budgets. Budgets are a way of keeping track of the factors that contribute to each characteristic of a spacecraft.

Pointing budgets quantify the effect of each error source on the overall vehicle pointing. Propellant budgets quantify the contribution of each maneuver involving thrusters to the consumption of fuel. On most spacecraft, propellant budgets are dominated by the fuel consumption to perform orbital maneuvers. Mass budgets provide information on the total mass and inertia matrix of a satellite. Power budgets determine overall spacecraft power consumption and are needed to size the power sources, usually solar arrays, and batteries.

9.2. Pointing budgets

There are several competing methodologies for doing pointing budgets for satellites. One of the most popular is to divide the error contributions into four temporal categories:

1. Bias (e.g., manufacturing misalignment that is fixed throughout the mission);
2. Long-term (e.g., seasonal variation in the yaw body fixed torque on a momentum-bias spacecraft without yaw sensing);
3. Diurnal (e.g., thermally induced rotation);
4. Short term (e.g., random noise);

and treat all contributions within each temporal category as if they were normally distributed uncorrelated random variables. The contributions within each category are then summed (by taking the square root of the sum of their squares). The totals for each temporary category are then added to get the final per-axis numbers.

Other methodologies examine each contribution and determine its nature, separating deterministic and other contributions from random. For example, orbit errors should be separated from alignment errors. This provides somewhat more conservative but statistically more justifiable results.

Take for example a geosynchronous communications satellite. This type of satellite points at the Earth for its entire mission life. The yaw axis is aligned with nadir, the pitch axis points South, and the roll axis points East. If the spacecraft has a single circular beam about nadir only roll and pitch errors matter. If it has off-nadir spot beams or if the main beam has a nonzero elevation, yaw couples into the beam errors. Once roll,

pitch, and yaw error values have been computed, they must be combined to form the beam-pointing accuracy. The beam is the radio-frequency wavefront that touches the planet. The beam-pointing accuracy is

$$\phi_{AZ} = \theta_{roll} + \frac{\delta_{AZ}}{180/\pi} \theta_{yaw} \quad (9.1)$$

$$\phi_{EL} = \theta_{pitch} + \frac{\delta_{EL}}{180/\pi} \theta_{yaw} \quad (9.2)$$

where δ_{EL} and δ_{AZ} are the beam-center offsets.

An example pointing budget is shown in Table 9.1. This is for the control mode using magnetic torquers and an Earth sensor. SA is the solar array. MWA is the

Table 9.1 Normal pointing mode pointing budget.

#	Item	Roll	Pitch	Yaw	Units
Bias					
1	ESA temperature effects	0.0150	0.0150	0.0000	deg
2	ESA optics w.r.t. cube	0.0250	0.0250	0.0000	deg
3	ESA cube to MRC	0.0250	0.0250	0.0000	deg
4	Antenna B.S. w.r.t. antenna RC	0.0180	0.0180	0.0000	deg
5	Antenna RC w.r.t. MRC	0.0150	0.01500	0.0000	deg
6	MWA spin axis w.r.t. mirror	0.0060	0.0060	0.0000	deg
7	MWA mirror w.r.t. MRC	0.0170	0.0170	0.0000	deg
8	Attitude sensing (Earth radiance)	0.0012	0.0012	0.0000	deg
9	S/C disturbance torque	0.0031	0.0076	0.0000	deg
	Subtotal	0.0486	0.0491	0.0000	deg
Diurnal					
10	East/West position	0.0100	0.0100	0.0000	deg
11	Attitude sensing (Earth radiance)	0.0100	0.0100	0.0000	deg
12	Orbit inclination	0.0100	0.0000	0.0100	deg
13	S/C disturbance torque	0.0080	0.0000	0.1470	deg
14	Antenna thermal distortion w.r.t. ESA RC	0.0100	0.0100	0.0000	deg
15	Thermal distortion MWA w.r.t. ESA RC	0.0200	0.0500	0.0000	deg
	Subtotal	0.0294	0.0529	0.1473	deg
Short Term					
16	Attitude sensor & actuator noise/resolution	0.0200	0.0200	0.0000	deg
17	SA stepping transients	0.0000	0.0500	0.0000	deg
18	S/C disturbance torques	0.0000	0.0000	0.0000	deg
	Subtotal	0.0200	0.0539	0.0000	deg
	Total	0.0980	0.1558	0.1473	deg

CEP = 0.1707 deg

momentum-wheel assembly. ESA is the Earth Sensor Assembly. S/C is spacecraft. MRC is the master reference cube that is the master alignment reference for the spacecraft. This is a cube that reflects laser light from a laser measuring device. CEP is the Circular Error Probability. The CEP in degrees says that for normally distributed random errors the 3-sigma probability is that the error will be within the CEP. This budget was generated for a momentum-bias spacecraft that uses an Earth sensor (ESA) to measure roll and pitch. The quantity of interest is the beam alignment (antenna boresight or B.S.) with respect to a target on the Earth. The temporal quantities are: bias, assumed fixed for the life of the spacecraft; diurnal, which vary with the orbit period; and short term, which changes at rates faster than orbit rate. Yaw is not sensed and is controlled solely by the spacecraft momentum. The largest component, in this case, is the orbit-rate disturbance so the only entries for yaw are orbit inclination and this disturbance-related error. In roll and pitch, there are many errors due to misalignments. Alignments are measured using an optical system. ESA RC is the reference cube for the Earth sensor.

ESA errors appear in all three temporal categories. These are taken from the manufacturer's specifications. All alignment errors are determined by measuring postvibration errors. The spacecraft is aligned and then placed on a shaker to simulate launch. The change in alignments is considered the alignment uncertainties.

9.3. Propellant budgets

Propellant budgets are detailed accounts of the propellant needed for all spacecraft operations. They include fuel needed for all orbit-change operations and for attitude control operations that may include pointing control, momentum unloading, reorientations, etc.

Attitude control fuel consumption can be divided into components that are proportional to the orbit-change fuel consumption and those that are not. The former is due to misalignments in the delta-V engines or asymmetries of multiple delta-V engines with respect to the center-of-mass. The attitude control system must cancel the disturbance torques due to the delta-V engines. The attitude control fuel requirement is usually found through simulation. A simulation can provide the torque and force demands for typical maneuvers. These are then used to compute the required pulsewidth demand and then the impulse that then leads to the fuel used per maneuver. For a blowdown propellant system, the thrust drops with time and this must be considered.

The second component includes firings that are not proportional to the magnitude of the delta-V. Both components must be determined through analysis and simulation. Orbit operations that involve changing the velocity of the spacecraft include:

1. Station acquisition;
2. Station keeping;
3. Station changes;

4. Deorbit.

Station keeping includes inplane and out-of-plane maneuvers. Station changes are generally within the plane of the orbit. Deorbit maneuvers include forcing an early reentry of the satellite for low-Earth orbit satellites and maneuvers to push geosynchronous satellites out of a geosynchronous orbit.

Table 9.2 shows an example propellant budget. This shows a budget for a hypothetical geosynchronous spacecraft through its second year of life. There are three thruster systems used. The mission-orbit system is monopropellant hydrazine and is divided into two half-systems that are pressurized independently. The third system is the solid rocket motor.

9.4. Power budgets

The attitude control system consumes power. The power consumption can be divided into standby power and active power. Standby power is the power consumed by a device when it is on but not active. For example, when a reaction wheel is enabled the power electronics will consume power. When the wheel is used for control the generation of torque will consume power. For a hydrazine monopropellant thruster, the catalyst-bed heaters consume power even if the thruster is not firing.

The active power consumed by the flight processors will be proportional to the processing load. You experience this with a laptop that can become uncomfortably hot when a computationally intense process is underway. Flight processors are shared so the attitude control power budget should only include the ACS contributions.

Different modes of operation will have different power needs. A power budget is needed for each mode. Each device should have a line listing standby, average, and maximum power consumption.

Maneuvers 1 through 4 prepare for the Apogee Kick Motor (AKM) firing. This rocket engine is used to put the spacecraft into geosynchronous orbit. They include a spin-precession maneuver (SPM) from negative orbit normal (NON) to the AKM burn attitude, a spin-up to the AKM spin rate (needed to stabilize the vehicle), and a small SPM trim. All use of the solid occurs during the AKM burn. This is followed by more attitude maneuvers to prepare for acquisition. Maneuvers 10 and 11 are drift-orbit maneuvers designed to get the spacecraft to the station. Usually, more than two are required. The remaining maneuvers are to correct East/West drift and inclination drift.

The inclination maneuvers use electrothermal hydrazine thrusters (EHT) so the specific impulses (the measure of fuel consumption per second) for delta-V are much higher. In all cases, the budget accounts for the different specific impulses for the attitude control thrusters and the delta-V thrusters.

Table 9.2 Example propellant budget.

Item	Description	Total	ACS Isp	DV Isp	Fuel Remaining	Fuel Remaining	Fuel Remaining	Fuel Used	Fuel Used	Fuel Used
1	Initial State	2055.92	0.00	0.00	117.89	117.89	865.00	0.000	0.000	0.000
2	SPM NON to AKM	2055.82	100.50	0.00	117.84	117.84	865.00	0.051	0.051	0.000
3	Spin Up 5 to 60 rpm	2051.14	118.33	0.00	115.50	115.50	865.00	2.341	2.341	0.000
4	SPM Trim	2051.13	100.10	0.00	115.50	115.50	865.00	0.001	0.001	0.000
5	AKM Firing	1188.56	0.00	265.00	115.50	115.50	2.43	0.000	0.000	862.574
6	Despin to 5 rpm	1183.88	118.33	0.00	113.02	113.30	2.43	2.481	2.202	0.000
7	SPM AKM to PON	1183.43	102.51	0.00	112.79	113.07	2.43	0.225	0.226	0.000
8	Despin to 2.53 rpm	1183.22	118.33	0.00	112.68	112.97	2.43	0.111	0.099	0.000
9	Despin from 1.1 rpm to orbit rate	1183.15	118.33	0.00	112.64	112.94	2.43	0.037	0.033	0.000
10	Drift-orbit station ac- quisition	1174.15	100.00	210.00	108.14	108.44	2.43	4.500	4.500	0.000
11	Drift-orbit inclina- tion correction	1163.94	100.02	360.00	103.05	103.32	2.43	5.093	5.115	0.000
12	East/West 1997	1163.89	100.00	143.01	103.03	103.30	2.43	0.022	0.022	0.000
13	East/West 1998	1163.85	100.00	143.02	103.00	103.28	2.43	0.022	0.022	0.000
14	North/South 1998	1159.97	100.02	360.00	101.07	101.33	2.43	1.936	1.943	0.000
15	East/West 1998	1159.92	100.00	144.29	101.05	101.31	2.43	0.022	0.022	0.000
16	East/West 1998	1159.88	100.00	144.30	101.02	101.29	2.43	0.022	0.022	0.000
17	East/West 1998	1159.84	100.00	144.31	101.00	101.27	2.43	0.022	0.022	0.000
18	North/South 1998	1155.97	100.02	360.00	99.07	99.33	2.43	1.930	1.936	0.000
19	East/West 1998	1155.93	100.00	145.57	99.05	99.31	2.43	0.022	0.022	0.000
20	East/West 1998	1155.88	100.00	145.58	99.03	99.29	2.43	0.022	0.022	0.000
21	East/West 1998	1155.84	100.00	145.60	99.01	99.27	2.43	0.022	0.022	0.000

9.5. Mass budgets

A mass budget includes the masses of all components of the vehicle and the inertias of all of those components. The budget includes the total of the masses and inertias. While masses are relatively easy to determine, inertias are not. Inertias can be calculated by using the mass distributions of components and the locations of those components

$$I = - \sum_{k=1}^N m_k (r_k - c)^\times (r_k - c)^\times \quad (9.3)$$

where r is the position vector of the component with respect to a reference, m_k is the mass of the component, and c is the position of the center-of-mass where

$$0 = \sum_{k=1}^N m_k (r_k - c) \quad (9.4)$$

This assumes that each mass can be approximated as a point mass. A better approximation is to enclose the component in a rectangular envelope and compute the inertia from the formula

$$I = \frac{m}{12} \begin{bmatrix} b^2 + c^2, 0, 0 \\ 0, a^2 + c^2, 0 \\ 0, 0, a^2 + b^2 \end{bmatrix} \quad (9.5)$$

where a is the x -length, b is the y -length, and c is the z -length. m is the total mass of the box. For larger components the inertia matrix can be computed on a test rig by applying 3-axis torques and measuring the resulting acceleration. Most CAD packages can also compute inertias.

When combining inertias to compute the total vehicle inertia it is necessary to know the alignments. Since alignment measurements have uncertainty, the best that will be known is the inertia to within the alignment errors.

One difficulty with creating a total spacecraft inertia matrix is that suppliers rarely provide an inertia matrix for a component. If the component is small its inertia matrix contribution is

$$I_k = -m_k d_k^\times d_k^\times \quad (9.6)$$

If a component is larger, this can be improved by computing the inertia matrix of a box that encloses the component. The mass of the component is then multiplied by the 3×3 inertia matrix for a uniform box.